

SYSTEM ADMINISTRATION

ENTERPRISE SECURITY

RELEASE V2.4

OCPP Charge Point Management System

Enterprise Administration, Multi-Tenancy, Granular RBAC, PKI Certificates, EPEX Tariffs, Auto-Heal & Live Protocol Diagnostics

TARGET AUDIENCE

System Administrators, Security Officers & Technical CPOs

SECURITY COMPLIANCE

OCPP 1.6 Security Profile 3 (mTLS) & ISO 15118

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August 2026

PUBLISHER

The Charge Grid

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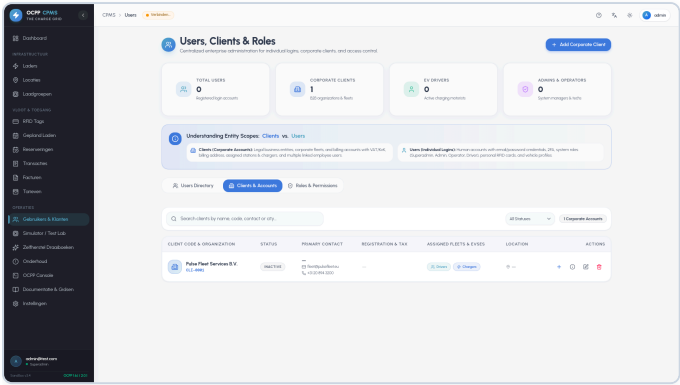
- 13. Scheduled Background Cron Jobs & Auto-Maintenance**

1. Multi-Tenant Architecture & Corporate Hierarchy

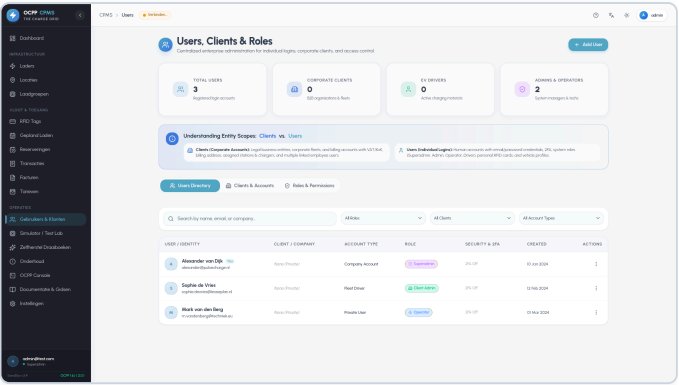
The CPMS architecture provides strict logical isolation between independent corporate clients, billing entities, and operational domains.

MULTI-TENANCY ISOLATION RULE

All database queries for non-superadmin users are strictly scoped by `companyId` or `owner_id`. Corporate client administrators can never view or modify chargers assigned to other organizations.



Corporate B2B Clients Directory



User Accounts & Roles Directory

2. Role-Based Access Control (RBAC Matrix)

The platform provides a granular 5-tier role hierarchy governing permissions across 6 system modules:

MODULE	SUPERADMIN	PLATFORM ADMIN	OPERATOR/TECH	CLIENT ADMIN	USER/DRIVER
Infrastructure & Stations	Full CRUD	Full CRUD	View & Diagnostics	Assigned Only	Map View
Remote Controls	Full Access	Full Access	Full Access	Restricted	Own Socket
Tariffs & Dynamic EPEX	Full CRUD	Full CRUD	View Only	View Assigned	View Rates
Invoices & SEPA Direct Debit	Full CRUD	Full CRUD	No Access	Own Invoices	Own Receipts
Roaming Hubs (OCPI/OICP)	Full CRUD	Manage	View Only	No Access	No Access
PKI Security & Audit Trail	Full CRUD	View Audit	No Access	No Access	No Access

OCPP CPMS
THE CHARGE GRID

Dashboard

INFRASTRUCTUUR

Laders

Locaties

Laadgroepen

VLOOT & TOEGANG

RFID Tags

Gepland Laden

Reserveringen

Transacties

Facturen

Tarieven

OPERATIES

Gebruikers & Klanten

Simulator / Test Lab

Zelfherstel Doorbekken

Onderhoud

OCPP Console

Documentatie & Gidsen

Instellingen

CPMS > Users Verbinden...

Users, Clients & Roles

Centralized enterprise administration for individual logins, corporate clients, and access control.

TOTAL USERS

0

Registered login accounts

CORPORATE CLIENTS

1

B2B organizations & fleets

EV DRIVERS

0

Active charging motorists

ADMINS & OPERATORS

0

System managers & techs

Understanding Entity Scopes: Clients vs. Users

Clients (Corporate Accounts):

Legal business entities, corporate fleets, and billing accounts with VAT/KVK, billing address, assigned stations & chargers, and multiple linked employee users.

Users (Individual Logins):

Human accounts with email/password credentials, 2FA, system roles (Superadmin, Admin, Operator, Driver), personal RFID cards, and vehicle profiles.

Users Directory

Clients & Accounts

Roles & Permissions

Role Capabilities & Permission Matrix

Granular system capabilities and functional access matrix across all CPMS modules.

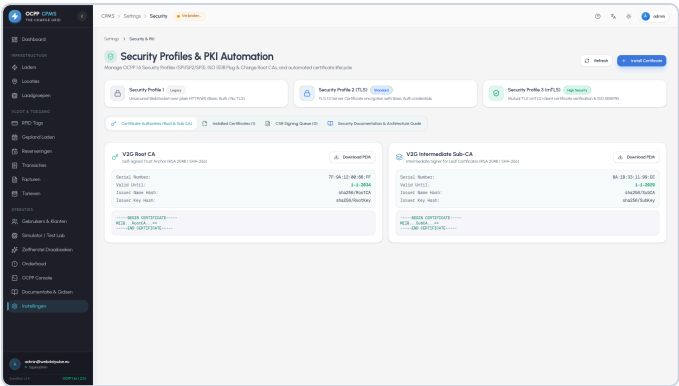
MODULE CAPABILITY	CATEGORY	SUPERADMIN	ADMIN	OPERATOR	CLIENT ADMIN	EV DRIVER
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Interactive Roles & Permissions Matrix Console

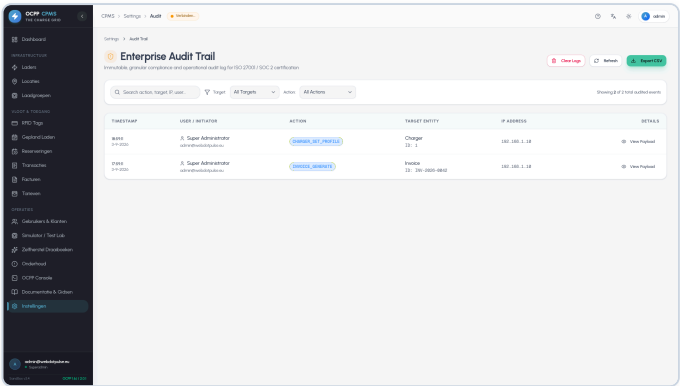
3. Security Profiles & PKI / TLS Certificates

The CPMS enforces standards compliant with **OCPP 1.6 Security Profile 3** and **ISO 15118 PKI**:

- **Profile 1:** Unencrypted HTTP / WS transport (restricted to private VPNs).
- **Profile 2:** TLS / WSS transport with HTTP Basic Authentication.
- **Profile 3:** Mutual TLS (mTLS) with bidirectional X.509 client and server certificates.
- **Certificate Authority Trust Store:** Upload and inspect Root CAs, Sub-CAs, and revocation lists.



PKI Security Profiles & X.509 Management



Enterprise Audit Trail & Compliance Log

4. Enterprise Audit Trail & Compliance

Every administrative action, remote control RPC invocation, and tariff modification is recorded immutably in the audit trail with UTC timestamp, actor IP, user-agent, target resource, and outcome status.

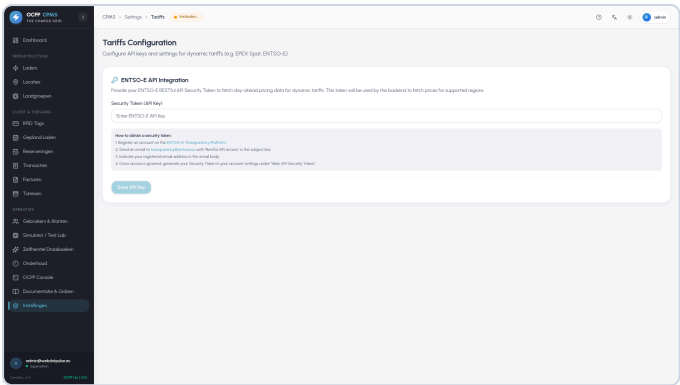
5. Dynamic EPEX Spot Pricing & Market Feeds

The **Dynamic Tariffs Engine** (`/settings/tariffs`) connects with wholesale Day-Ahead electricity markets to compute real-time hourly consumer tariffs:

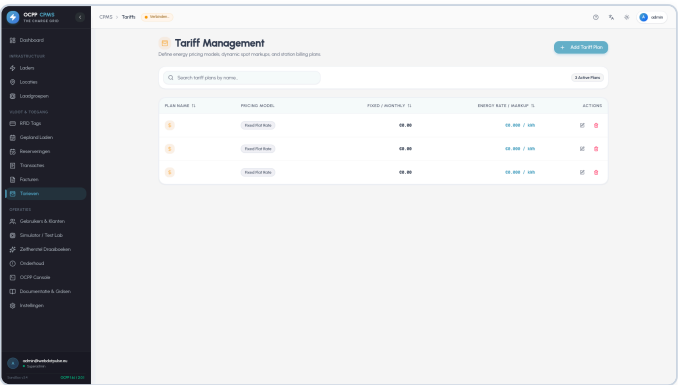
SPOT PRICING FORMULA

$$\text{Tariff (£/kWh)} = (\text{Spot Price} \times \text{Multiplier}) + \text{CPD Markup (£/kWh)} + \text{Grid Fee} + \text{VAT (21\%)}$$

- **Supported Market Feeds:** EnergyZero API, ENTSO-E Transparency Platform, and Energy-Charts API.
- **Automated Cron:** Daily price ingestion fires at 13:15 CET for the subsequent 24-hour delivery period.



Dynamic EPEX Spot Tariffs Settings



Tariffs Overview Matrix

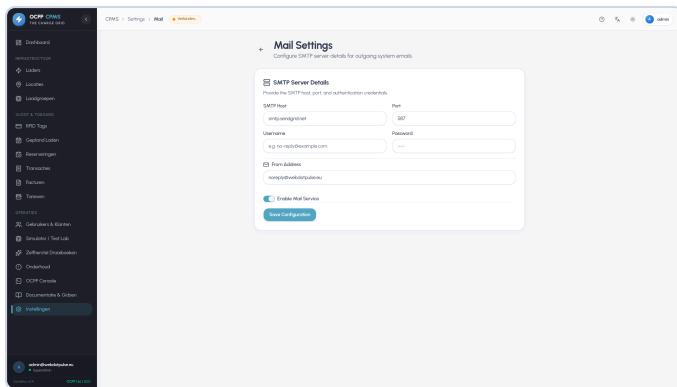
6. SMTP Mail Server & HTML Template Engine

6.1 Outgoing SMTP Configuration (/settings/mail)

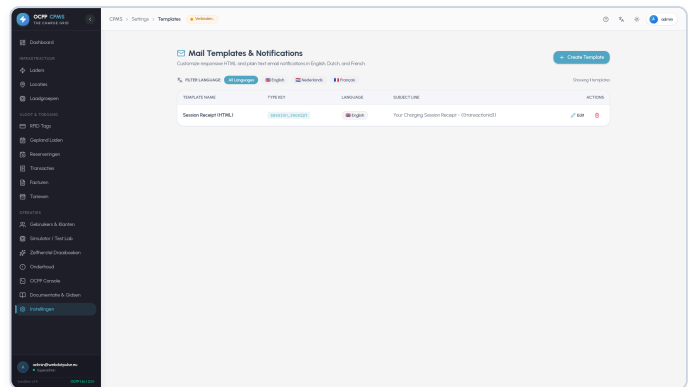
Configure transactional email delivery via Postmark, SendGrid, Amazon SES, or custom SMTP relays with TLS encryption and testing tools.

6.2 HTML Mail Template Editor (/settings/templates)

Customize email templates with visual previews and dynamic variables: `{{user_name}}` , `{{invoice_number}}` , `{{total_amount}}` , `{{kwh_delivered}}` .



SMTP Outgoing Mail Server Configuration

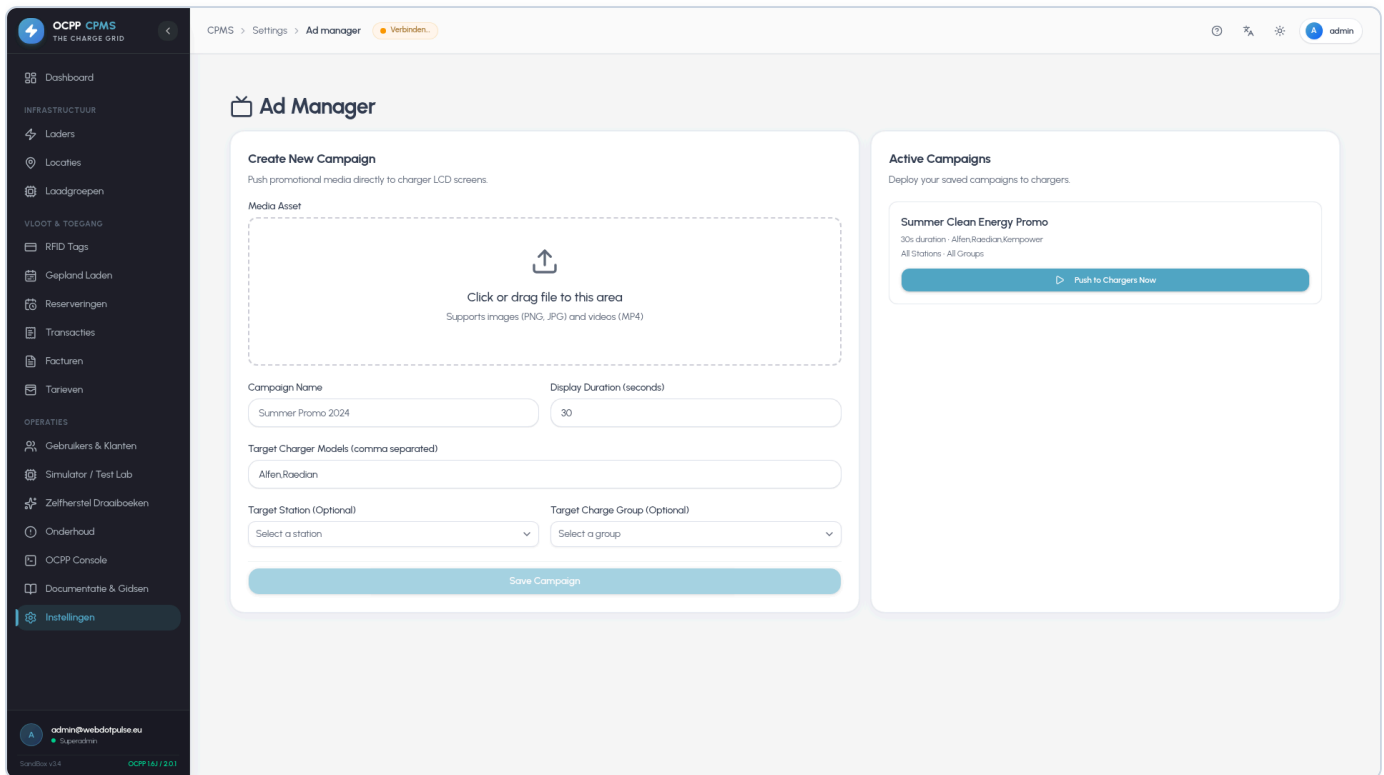


HTML Mail Template Visual Editor

7. Screen Advertising Manager & Media Campaigns

Distribute digital advertisement media campaigns to chargers equipped with color LCD screens:

- Supports MP4 (H.264 video), PNG, JPG, and dynamic HTML5 promo banners.
- Target campaigns by specific stations, charge groups, or geographic cities.
- Distribute assets via customized OCPP `DataTransfer` vendor RPCs.

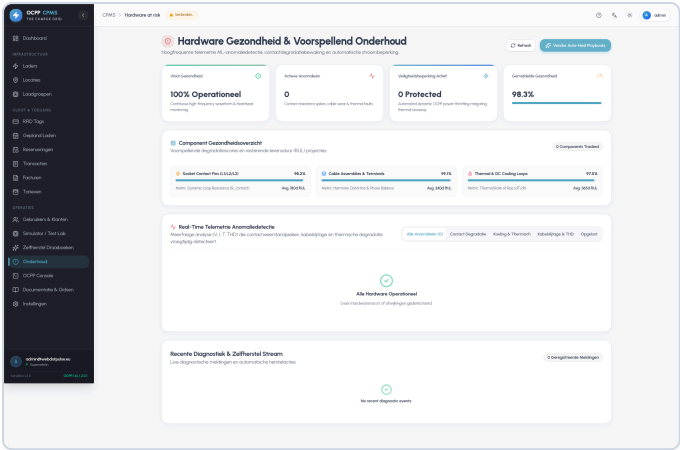


Screen Advertising Campaign Scheduler

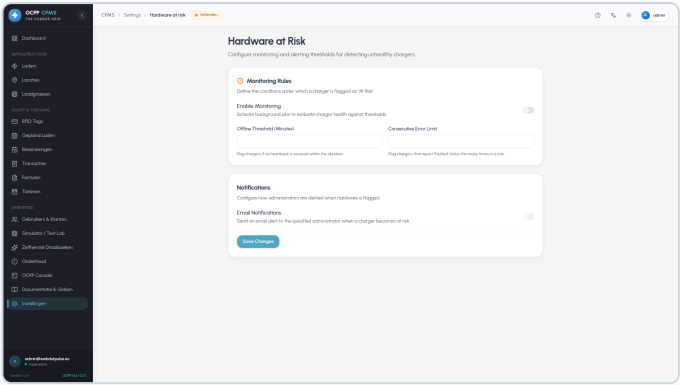
8. Hardware-at-Risk Engine & Auto-Heal Rules

The **Hardware-at-Risk Subsystem** (`/hardware-at-risk` & `/settings/hardware-at-risk`) continuously evaluates telemetry to resolve hardware failures automatically:

AUTO-HEAL PROTOCOL
If a charger misses heartbeats for >180 seconds or experiences a suspended EVSE lock, the engine automatically issues a remote soft reset before escalating an incident ticket to field technicians.



Hardware-at-Risk Live Fleet Monitor



Auto-Heal Heuristic Rules Configuration

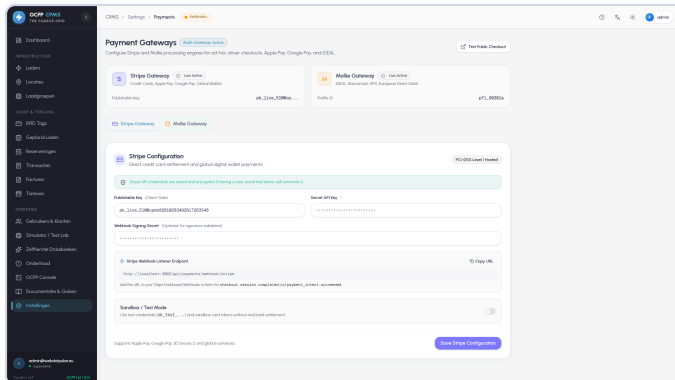
9. Payment Gateways & Roaming Connections

9.1 Stripe & Mollie Gateway Setup (/settings/payments)

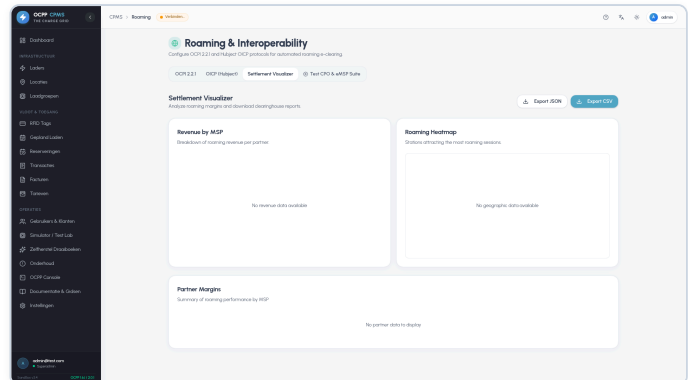
Manage API credentials, test sandbox modes, and webhook signing secrets for instant public ad-hoc payment processing.

9.2 Roaming Hubs (OCPI 2.2.1 & Hubject OICP)

Manage bilateral roaming credentials, exchange tokens, and track wholesale clearinghouse settlements.



Mollie & Stripe Payment Gateway Credentials



Roaming Settlement & Clearinghouse Visualizer

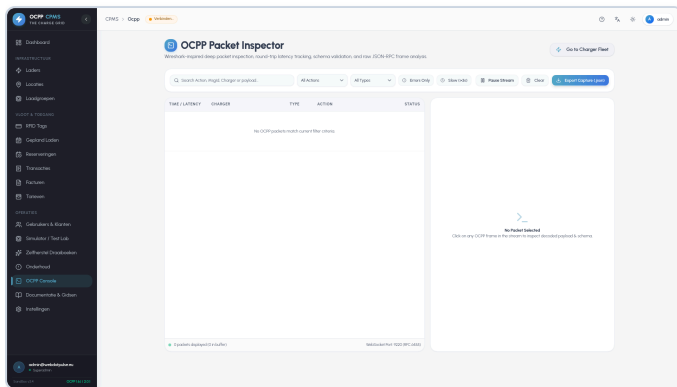
10. Live OCPP Packet Inspector & Quirk Profiles

10.1 Live Packet Inspector (/ocpp)

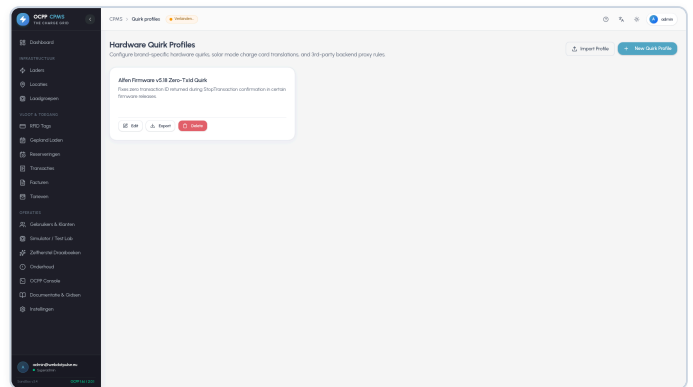
Stream real-time unbuffered WebSocket JSON-RPC frames (`CALL` , `CALLRESULT` , `CALLERROR`) with full syntax parsing and manual RPC dispatcher.

10.2 Hardware Quirk Profiles (/quirk-profiles)

Non-intrusively repair vendor-specific OCPP firmware non-compliance (missing power derivation, integer multipliers, UID endianness).



Live OCPP Packet Inspector & RPC Console



Hardware Quirk Profiles & Overrides

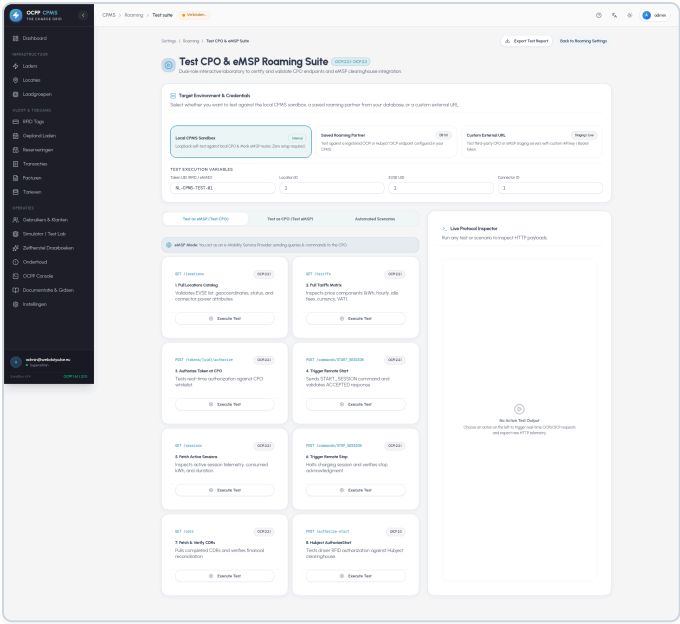
11. Roaming Test Suite & Simulator Studio

11.1 Roaming Conformance Test Suite (/roaming/test-suite)

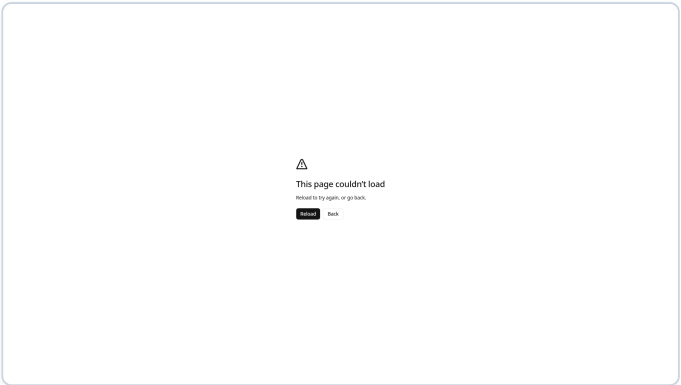
Automate validation of eMSP and CPO partner endpoints against synthetic charging sessions and CDR schemas before production activation.

11.2 Digital Twin Charger Simulator Studio (/simulator)

Simulate realistic multi-protocol charging station hardware, connector states, and fault scenarios directly in the browser.



Roaming Conformance Test Suite



Charger Simulator Studio

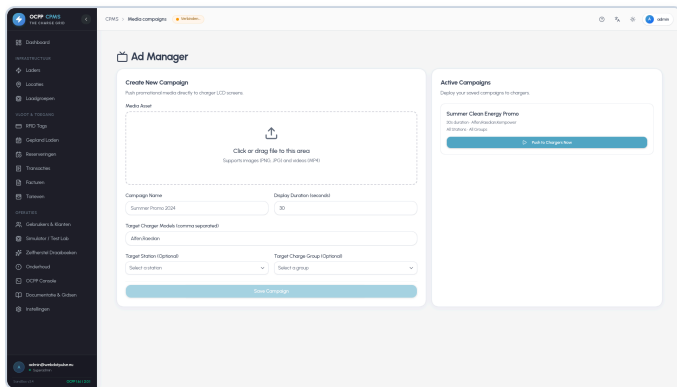
12. Media Screen Campaigns & Firmware Management

12.1 Media Campaigns Scheduler (/media-campaigns)

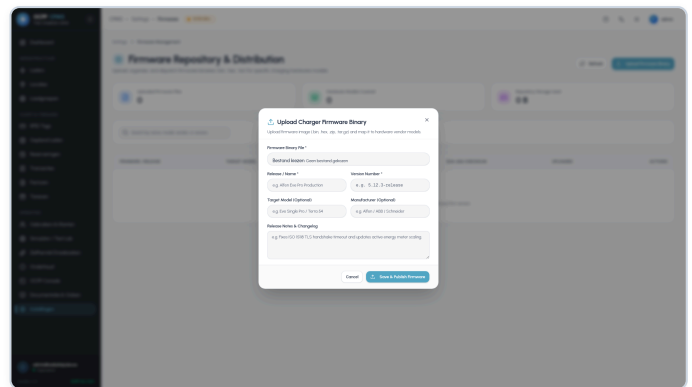
Schedule high-definition video and promotional image campaigns onto digital display screens of modern EVSE hardware.

12.2 Firmware Binary Over-The-Air (OTA) Updates (/settings/firmware)

Centrally upload and deploy vendor firmware binaries across the fleet via standard OCPP `UpdateFirmware` RPC commands.



Media Campaigns Scheduler



Firmware Binary Management & OTA Deployment